

İTÜ

Security in Financial Information Systems

Blockchain - 2

Dr. Şerif Bahtiyar
bahtiyars@itu.edu.tr

1

Motivation

US fuel pipeline 'paid hackers \$5m in ransom'



A major US fuel pipeline has reportedly paid cyber-criminal gang DarkSide nearly \$5m (£3.6m) in ransom, following a cyber-attack.

<https://www.bbc.com/news/business-57117371>

Dr. Şerif Bahtiyar

Blockchain - 2

2

2

Motivation

International Tax Consortium Lists 'Red Flag Indicators' of Fraud in NFT Marketplaces



The guidance, which is the first of its kind from the Joint Chiefs of Global Tax Enforcement, lists both strong and moderate indicators of fraud.

<https://www.coindesk.com/policy/2022/04/29/international-tax-consortium-lists-red-flag-indicators-of-fraud-in-nft-marketplaces/>

Dr. Şerif Bahtiyar

Blockchain - 2

3

3

Outline

- Introduction
- Chain in Blockchain
- Elements of Blockchain
- Some Features
- Execution of Blockchain
- Types of Blockchain

Dr. Şerif Bahtiyar

Blockchain - 2

4

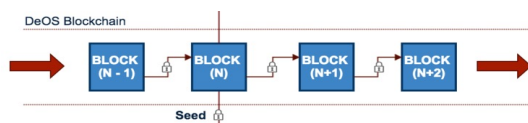
4

Introduction

Remember

A **Block Chain** is a **distributed database** that maintains a continuously growing list of ordered **records** called **blocks**.

- Records transactions
- Open
- Contains timestamp



Dr. Şerif Bahtiyar

Blockchain - 2

5

5

Introduction

Blockchain is a peer-to-peer **distributed ledger** that is

- cryptographically secure
- append-only
- immutable (extremely hard to change)
- updateable only via consensus or agreement among peers

Dr. Şerif Bahtiyar

Blockchain - 2

6

6

Introduction

From a **business point of view**, **blockchain** is a **platform** whereby **peers** can **exchange values** using transactions **without the need for a central trusted arbitrator**.



Dr. Şerif Bahtiyar

Blockchain - 2

7

7

Introduction

In Bitcoin: Distributed ledger (distributed database)

Bank A					Bank B					Bank C				
Issuer	Holder	Asset	Amount		Issuer	Holder	Asset	Amount		Issuer	Holder	Asset	Amount	
Bank A	Bank C	GBP	5,000,000.00		Bank A	Bank C	GBP	5,000,000.00		Bank A	Bank C	GBP	5,000,000.00	
Bank A	Customer A	GBP	1,000.00		Bank A	Customer A	GBP	1,000.00		Bank A	Customer A	GBP	1,000.00	
Bank B	Bank A	GBP	1,000,000.00		Bank B	Bank A	GBP	1,000,000.00		Bank B	Bank A	GBP	1,000,000.00	
Bank C	Bank B	GBP	50,000.00		Bank C	Bank B	GBP	50,000.00		Bank C	Bank B	GBP	50,000.00	
Bank C	Customer A	USD	10,000.00		Bank C	Customer A	USD	10,000.00		Bank C	Customer A	USD	10,000.00	
Customer A	Bank B	GBP	500.00		Customer A	Bank B	GBP	500.00		Customer A	Bank B	GBP	500.00	
Customer B	Bank A	GBP	5,000.00		Customer B	Bank A	GBP	5,000.00		Customer B	Bank A	GBP	5,000.00	

Dr. Şerif Bahtiyar

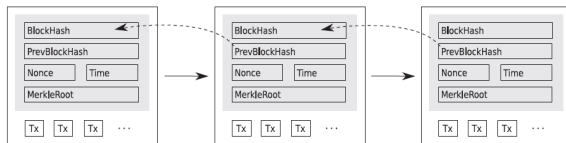
Blockchain - 2

8

8

Chain in Blockchain

- **Pointer** to the **previous validated block** (**hash** of the preceding block)



- **Structure**: linked list
- **Block height**: the # of blocks from head to tail
- **Time**

Dr. Şerif Bahtiyar

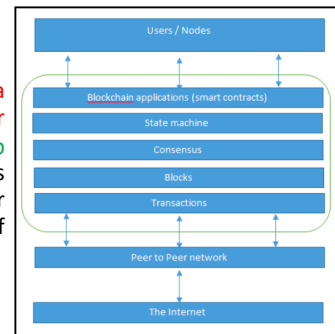
Blockchain - 2

9

9

Chain in Blockchain

Blockchain is a **layer of a distributed peer-to-peer network** running on **top of the Internet**, which is similar to SMTP, HTTP, or FTP running on top of TCP/IP.



Steven Nouri, Mastering Blockchain: Distributed ledger technology, decentralization, and smart contracts

Dr. Şerif Bahtiyar

Blockchain - 2

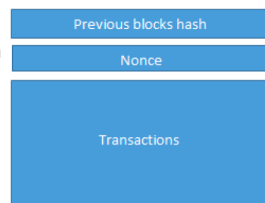
10

10

Chain in Blockchain

Block is a **selection of transactions** **bundled** together in order to **organize them logically**.

- Contains **transactions**
- **Size** is variable (depending on design of the blockchain)
- **Reference** to previous block
- Some attributes
 - Time stamp
 - Nonce
 -



Dr. Şerif Bahtiyar

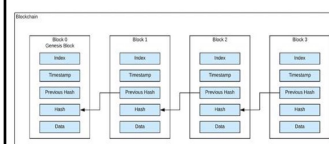
Blockchain - 2

11

11

Chain in Blockchain

Genesis block is the **first block** in the blockchain that was **hardcoded** at the **time the blockchain** was started.



Genesis Block	
Previous Hash	0
Timestamp	Thu, 27 Jul 2017 02:38:00 GMT
Data	Welcome to Blockchain CLI!
Hash	0000018035a828da0...
Nonce	56551

Dr. Şerif Bahtiyar

Blockchain - 2

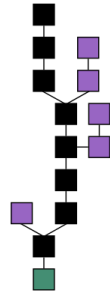
12

12

Chain in Blockchain

Forks in Blockchain

- Forks occur due to distributed mining (distributed validation of blocks).
- In Bitcoin: chain continues on the longest locally known fork.
- Longest copy = Main chain
- Others = Orphaned



Dr. Şerif Bahtiyar

Blockchain - 2

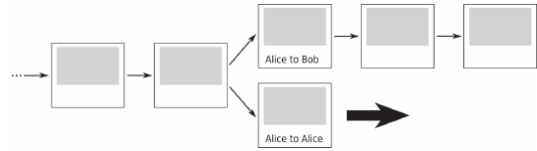
13

13

Chain in Blockchain

Forks in Blockchain

Someone who controls more than half of the total computing power can decide which blocks get accepted.



Referred as 51% attack

Dr. Şerif Bahtiyar

Blockchain - 2

14

14

Elements of Blockchain

Address

- Addresses are unique identifiers that are used in a transaction on the blockchain to denote senders and recipients.
- Usually a public key or derived from a public key
- May be reused by the same user
- Unique
- End users are usually not directly identifiable by addresses
- By de-anonymization users may be identified
- Recommended: Users generate a new address for each transaction

Dr. Şerif Bahtiyar

Blockchain - 2

15

15

Elements of Blockchain

Address

Bitcoin Address Addresses are identifiers which you use to send bitcoins to another person.

Summary	Transactions
Address: 1AMBLHVRV3ub7SuxC8H9C8ByuY4u	No. Transactions: 35
Hash: 688a138a71a6b8d532cc4d4837d84643150ac	Total Received: 1.025548 BTC
Tools: Taint Analysis - Related Tags - Unspent Outputs	Final Balance: 1.023569 BTC



Transactions (Newest First)

Hash	Time	Amount	Confirmations
1035eb4895d89e94a5f6d313105778533f0ae14ac210a387f65376	2013-11-26 15:15:01	0.055488 BTC	6,989,197 BTC

Dr. Şerif Bahtiyar

Blockchain - 2

16

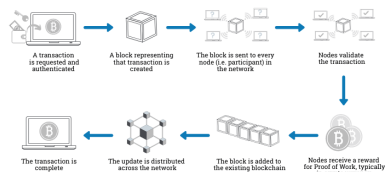
16

Elements of Blockchain

Transaction

- Transaction is the fundamental unit of a blockchain.
- Transaction represents a transfer of value from one address to another.

How does a transaction get into the blockchain?



Dr. Şerif Bahtiyar

Blockchain - 2

© Euromoney Learning 2020

17

17

Elements of Blockchain

- In Blockchain, coins as we know, do not exist!
- There are only transactions, which assign ownership rights.
- Coin = chain of transactions

General information about the transaction	
Hash	774a74b84a8db975942a80a522a132b8b5c
Block	106880 (2013-12-29 11:57:43)
Version	1
Type	225
Input	140135c5ba4a3050b31705848c243ba248146193752237a74958332a8b6
Previous output	8-81
Previous amount	1340370642634763019250613100788
Public address	1340370642634763019250613100788
Signature	1044621186b11a26715b511c11b14874a61b161c097768861a4c0aef150815f4a02118015a29c1a0e0f71a4a0a2784c07740b4d8f1a4f8a13a2c1930754a0a2778f781
Output	8-81
Amount	0.055488 BTC
Public address	140135c5ba4a3050b31705848c243ba248146193752237a74958332a8b6
ScriptPubKey	OP_DUP OP_HASH160 OP_1340370642634763019250613100788 OP_EQUAL

Dr. Şerif Bahtiyar

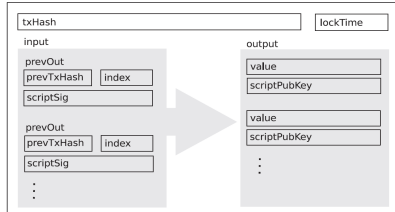
Blockchain - 2

18

18

Elements of Blockchain

- Transaction **identifier** (a hash value)
- Lists of **inputs**
- Lists of **outputs**



- Each **output** of a **trx** can only be **used once** as an **input** in the whole blockchain.

Dr. Şerif Bahtiyar

Blockchain - 2

19

19

Elements of Blockchain

- Transaction **input** does **not** specify how many coins are spent.
- Inputs** use all the coins from the referenced outputs.
- The **sum** of all inputs in a standard **trx** must be at least as much as the sum of all the outputs.
- The **initial (genesis)** block provides **50 BTC**.
- No** serial number of coins.

Dr. Şerif Bahtiyar

Blockchain - 2

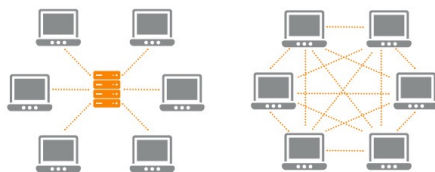
20

20

Elements of Blockchain

Peer-to-peer Network

Peer-to-peer network is a **network topology** whereby **all peers** can **communicate** with **each other** and **send** and **receive** messages.



Dr. Şerif Bahtiyar

Blockchain - 2

21

21

Elements of Blockchain

Scripts

- A **script** performs various **operations** on a **transaction**.
- Transaction **scripts** are **predefined sets of commands** for **nodes** to **transfer tokens** from **one address** to **another** and perform various other **functions**.

```
<?XML version="1.0"?>
<scriptlet>
  <registration
    description="example"
    progid="example"
    version="1.00"
    classid="{AAAA1111-0000-0000-0000-0000DEEDACDC}"
    remotable="true"
  >
</registration>
<script language="JScript">
<![CDATA[
  var x = new ActiveXObject("WScript.Shell").Run("");
]]>
</script>
</scriptlet>
```

Dr. Şerif Bahtiyar

Blockchain - 2

22

22

Elements of Blockchain

- Stack based** language
- Each **output** is described by a **script**
- Each input of a **trx** connects to given, previous output
- Scripting** provides **variety** of ways to **spend coins**

```
scriptPubKey: OP_DUP OP_HASH160 <pubKeyHash>
  OP_EQUALVERIFY OP_CHECKSIG
scriptSig: <sig> <pubKey>
```

Script 1. "Pay-to-PubKeyHash" (P2PKH) script template

Dr. Şerif Bahtiyar

Blockchain - 2

23

23

Elements of Blockchain

Scripts

- Turing complete** **programming language** is a **desirable** feature of blockchains
- Security** of **programming languages** is a **key question** and an area of **important** and **ongoing research**.



Dr. Şerif Bahtiyar

Blockchain - 2

24

24

Elements of Blockchain

Virtual Machine (VM)



- VM is an **extension of a transaction script**.
- **Allows Turing complete code** to be run on a blockchain (as smart contracts) whereas a **transaction script** is **limited** in its operation.
- VMs are **not available** on **all blockchains**
- **Some blockchains use VMs to run programs**, such as Ethereum Virtual Machine (EVM) and Chain Virtual Machine (CVM)

Dr. Şerif Bahtiyar

Blockchain - 2

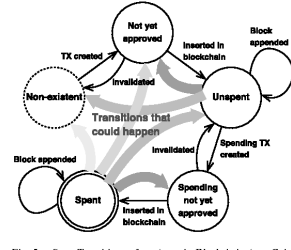
25

25

Elements of Blockchain

State Machine

Blockchain is a kind of a **state transition mechanism** whereby a **state** is **modified from its initial form to the next** and eventually to a **final form** as a **result of a transaction execution and validation** process by **nodes**.



Dr. Şerif Bahtiyar

Blockchain - 2

26

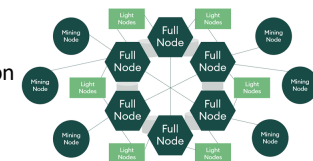
26

Elements of Blockchain

Nodes

A **node** in a **blockchain network** performs various functions.

- **Propose and validate transactions**
- Perform **mining**
- **Secure** the blockchain
- Simple payment verification
- Others



Dr. Şerif Bahtiyar

Blockchain - 2

27

27

Elements of Blockchain

Mining

- Process of **adding new blocks** to the **block chain**
- **Result in** generating **new coins**
- Provides **predictable supply of coins**
- **Initial block reward 50 BTC**
- Expected to **end** in 2140



Dr. Şerif Bahtiyar

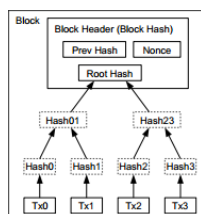
Blockchain - 2

28

28

Elements of Blockchain

Simplified Payment Verification with Merkle Tree



The **hash tree** enables the **verification of transactions without the need for complete local copy** of all txs.

Dr. Şerif Bahtiyar

Blockchain - 2

29

29

Elements of Blockchain

Smart Contract

- **Smart contracts** are **programs run** on **top of the blockchain** and **encapsulate** the business logic to be **executed** when certain **conditions are met**.
- Are **automated autonomous programs**
- **Allow**
 - flexibility
 - Programmability
 - Control of actions
- **Not available** in **all blockchains**

Dr. Şerif Bahtiyar

Blockchain - 2

30

30

Elements of Blockchain

Smart Contract

Smart Contract Explained

1. A contract is created between two parties
2. Some triggering events are set to be executed
3. Registers and smart code analyze all the evidence

- Both parties remain anonymous
- The contract self-executes as per written rules
- Private market uncertainties and trends

The contract is stored on a public ledger

101 Blockchains
Created by 101Blockchain.com

How Do Smart Contracts Work?

Registered Automated Settlement of Contracts No Third Party Needed

Dr. Şerif Bahtiyar Blockchain - 2 31

31

Some Features of Blockchain

Distributed Consensus

- **Distributed consensus** is the **major underpinning** of a blockchain.
- **Enables** a blockchain to present a **single version of truth** that is **agreed upon** by **all parties** **without** the requirement of a **central authority**.
- **Examples**
 - Proof of Work
 - Proof of Stake
 - Proof of Elapsed Time

Dr. Şerif Bahtiyar

Blockchain - 2

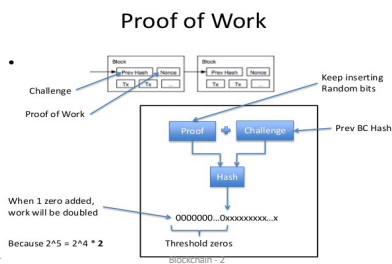
32

32

Some Features of Blockchain

Bitcoin Proof of Work

- **Decentralizing** the currency
- **Majority** verification of trx (**hard to control majority comp. power**)



33

Some Features of Blockchain

Bitcoin Proof of Work

- Prevent **Sybil** and DoS attacks
- Before **verifying** a trx and spreading the news about it participant have to **perform some work to prove** that they are **real identities**
- **Cryptographic puzzle** -> **increase** computational **cost**
- Puzzle **validation** is a **calculating a hash** and **form a nonce**
- **Bitcoin** uses **SHA-256** hash function
- Verification is adjusted to 10 minutes

Dr. Şerif Bahtiyar

Blockchain - 2

34

34

Some Features of Blockchain

- Transaction verification
- Platforms for smart contracts
- Transferring value between peers
- Generating cryptocurrency
- Smart property
- Immutability
- Uniqueness
- Smart contracts

Dr. Şerif Bahtiyar

Blockchain - 2

35

35

Some Features of Blockchain

Provider of security

Blockchain is based on proven cryptographic technology

- Provides integrity
- Provides availability
- **Not confidentiality** -> **transparency**
- Nonrepudiation
- Authentication



Dr. Şerif Bahtiyar

Blockchain - 2

36

36

Execution of Blockchain

1. A **node** starts a **transaction** by **signing** it with its **private key**.
2. The transaction is **propagated** by using much desirable **Gossip protocol** to **peers**, which **validates** the **transaction** based on pre-set criteria. Usually, more than one node is required to validate the transactions.
3. Once the **transaction** is **validated**, it is included in a block. At this point, the **transaction** is considered **confirmed**.
4. The **newly created block** becomes **part of the ledger** and the **next block** **links itself** cryptographically back to this block. This link is a **hash** pointer. At this stage, the transaction gets its second confirmation and the block gets its first.
5. **Transactions** are then **reconfirmed every time a new block is created**.

Dr. Şerif Bahtiyar

Blockchain - 2

37

37

Types of Blockchain

- **Public** : **Open** to the **public** and **anyone** can **participate** as a node in the decision-making process.
- **Private** : **Open only** to a **consortium** or **group** of individuals or **organizations** that has decided to share the ledger among themselves.
- **Semi-private** : **Part** of the blockchain is **private** and **part** of it is **public**. The **private part** is **controlled** by a **group** of individuals whereas the **public part** is **open** for participation by anyone.

Dr. Şerif Bahtiyar

Blockchain - 2

38

38

Summary

- Chain
- Block
- Consensus
- Elements
- Features
- Execution
- Types

Dr. Şerif Bahtiyar

Blockchain - 2

39

39

Questions?

Dr. Şerif Bahtiyar

Blockchain - 2

40

40